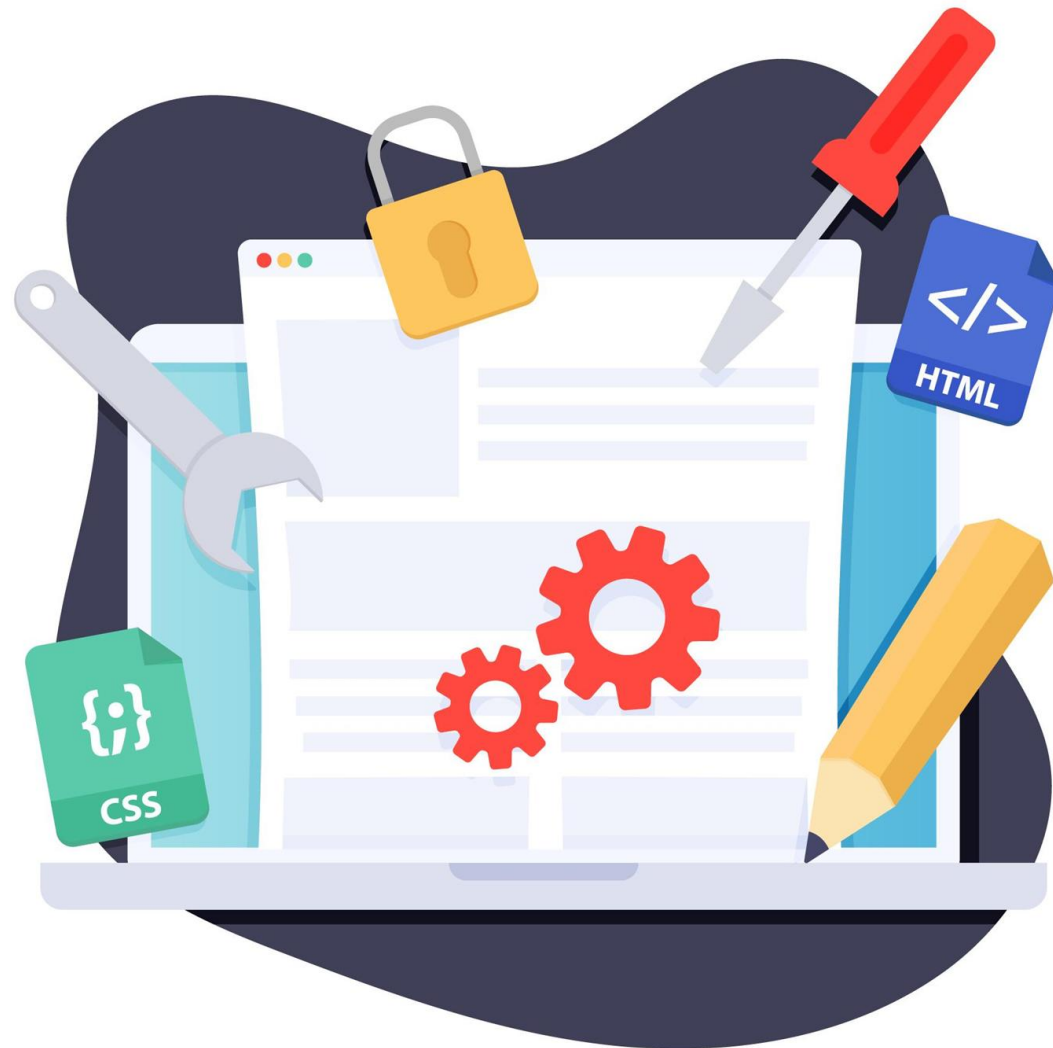


Python Project for Data Science

17/04/2024

Web Scrapping



Course Overview

You will perform specific data science and data analytics tasks such as extracting data, web scraping, visualizing data and creating a dashboard. This project will showcase your proficiency with Python and using libraries such as Pandas and Beautiful Soup within a Jupyter Notebook. Upon completion you will have an impressive project to add to your job portfolio.

Session Content

- What is Web Scrapping?
- How do we do?
- Tools to use
- Ethics for scraping
- Demo



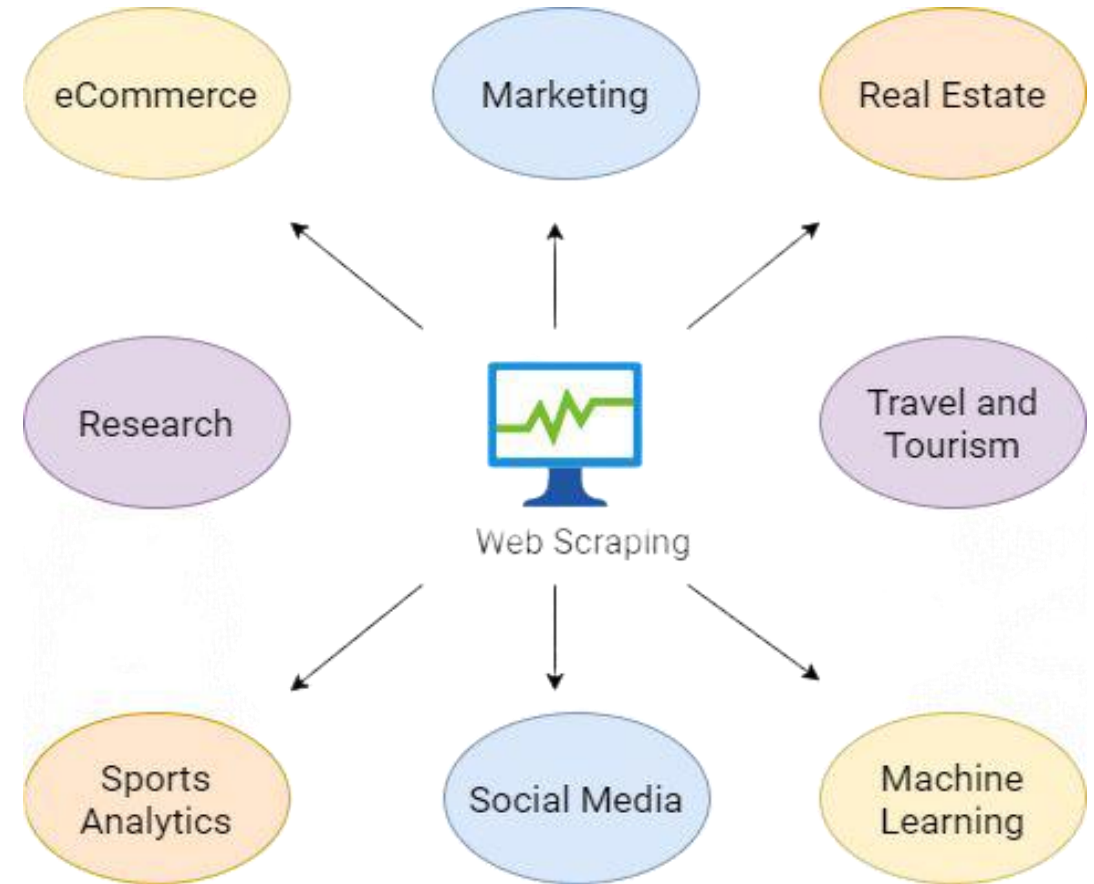
What is Web Scrapping?

- **Web scraping:** is technique for gathering data or information on web pages.
- **Web scraping:** is method to extract data from a website that does not have an API, or we want to extract LOT of data which we can not do through an API due to rate limiting.
- Through web scraping we can extract any data which we can see while browsing the web.
- You could revisit your favorite website every time it updates for new information, Or you could write a web scraper to have it



Web Scraping in Real Life

- Extract products information
- Extract job posting and internships
- Extract offers and discount from deal of the day website
- Extract data to make search engine
- Gathering weather data
- Etc.



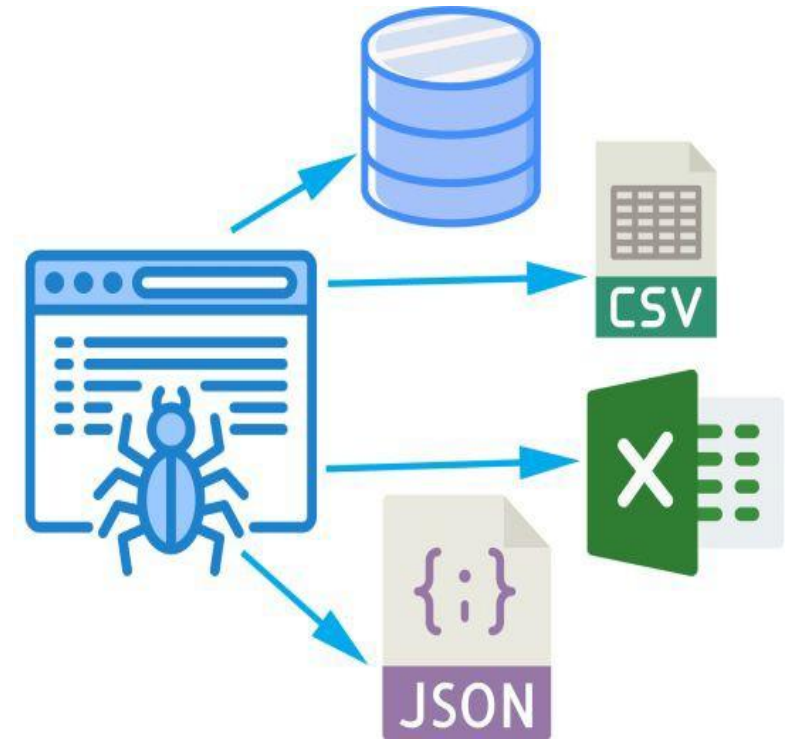
Advanced Web Scrapping Vs. API

- Web scrapping is not rate limited
- Anonymously access the website and gather data
- Some website don't have API
- Some data is not accessible through an API



Workflow

- Web scraping follows this workflow:
- Get the website – using HTTP library
- Parse the html document – using any parsing library
- Store the results – either a db , csv, txt file etc.



Libraries

- BeautifulSoup
(bs4)
- Lxml
- Selenium
- Re
- scrapy



Is Web Scraping Legal?

In short, the action of web scraping is not illegal. However, some rules need to be followed. Web scraping is illegal when non-publicly available data is extracted.

Demo 1

Beautiful Soup

Demo 2

Selenium

Questions & Answers



Thank you!